

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTETC401 - NETWORK THEORY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

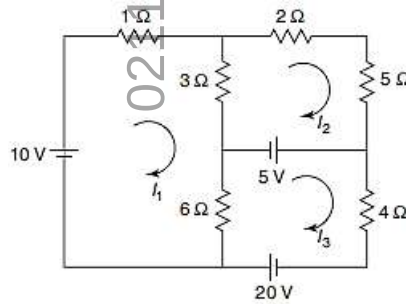
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- 1 In nodal analysis, what is the quantity solved at each essential node?
A) Current B) Voltage C) Resistance D) Power
- 2 The Norton equivalent of a circuit is a:
A) Voltage source in series with a resistor B) Current source in series with a resistor
C) Current source in parallel with a resistor D) Voltage source in parallel with a resistor
- 3 A tree in a graph is a subgraph that:
A) Includes all loops B) Includes all nodes and no loops
C) Excludes all nodes D) Contains only branches with resistors
- 4 In a source-free RL circuit, the current decays:
A) Exponentially B) Linearly C) Logarithmically D) Sinusoidally
- 5 Which of the following is a valid assumption for circuit analysis?
A) All components are ideal B) The voltage drop across all components is negligible
C) The power loss in components is infinite D) The current in all components is constant
- 6 Maximum power transfer occurs when the load resistance is:
A) Greater than the source resistance B) Equal to the source resistance
C) Less than the source resistance D) Independent of the source resistance
- 7 The Laplace transform of a unit step function $u(t)$ is:
A) 1 B) $1/s$ C) s D) $1/(s+1)$
- 8 Which component contributes an initial condition in s-domain circuit analysis?
A) Resistor B) Inductor and Capacitor C) Voltage source D) Current source
- 9 Z-parameters are also known as:
A) Admittance parameters B) Transmission parameters
C) Impedance parameters D) Hybrid parameters
- 10 Which parameters are most suitable for analyzing cascaded two-port networks?
A) Z-parameters B) Y-parameters C) h-parameters D) ABCD parameters

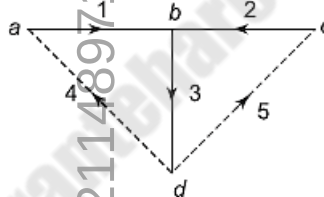
- 11 Network synthesis involves:
- Only frequency domain analysis
 - Finding transfer functions from time responses
 - Designing a network that realizes a given impedance or transfer function
 - Analyzing power in an existing network
- 12 A first-order low-pass RC filter passes:
- High frequencies
 - All frequencies
 - Only low frequencies and attenuates high frequencies
 - Only DC

Q2. Solve the following.

- A) State Kirchhoff's laws and Find the current through the $6\ \Omega$ resistor is shown in Figure

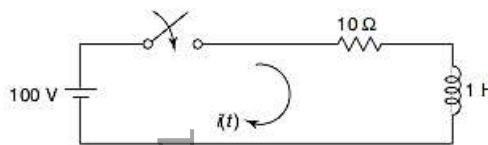


- B) Define the term: Rank of a Graph, Tree. Also Obtain Tieset and f-cutset matrix for the following graph.

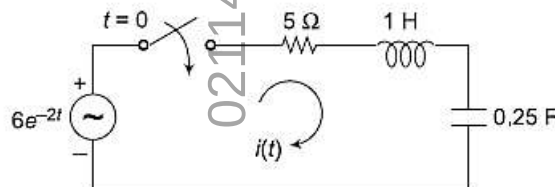


Q3. Solve the following.

- A) In the given network of Fig. shown, the switch is closed at $t = 0$. With zero current in the inductor, find i , di/dt , and d^2i/dt^2 at $t = 0^+$



- B) The switch is closed at $t = 0$. Determine current $i(t)$, assuming zero initial conditions, using Laplace transform



Q4. Solve Any Two of the following.

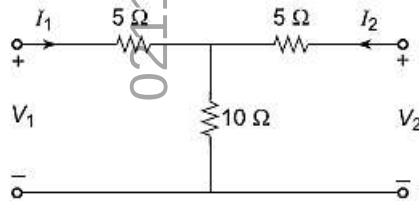
- A) Explain Circuit elements in the s-domain

- B) Write the Laplace transform of the following standard Network Functions. 6
- Unit Step Function
 - Exponential Function
 - Sine Function
 - Cosine Function
 - Ramp Function
 - Delayed Ramp

- C) Derive an Expression for response of RC circuit, When the unit step signal is applied. 6

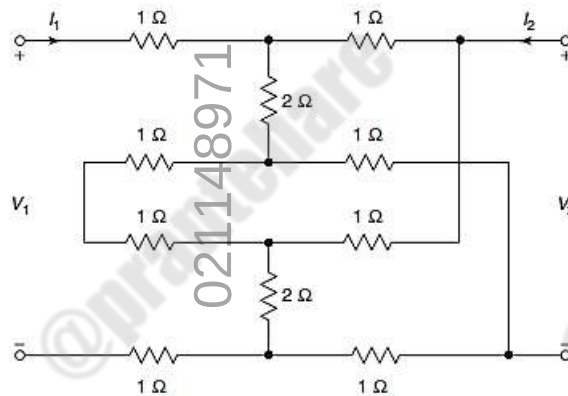
Q5. Solve Any Two of the following.

- A) Find Z parameters of the following two port network, Comment on Symmetry and Reciprocity. 6



- B) Derive Y-parameter in terms of h-parameter and ABCD parameter 6

- C) Determine h-parameters for the network shown 6



Q6. Solve Any Two of the following.

- A) Check for the Hurwitz polynomial 6

- $s^5 + s^3 + s$
- $s^4 + 6s^3 + 8s^2 + 10$

- B) Realise the following function in Foster-I and Foster-II forms 6

$$Z(s) = \frac{3(s+2)(s+4)}{s(s+3)}$$

- C) Explain about classification of filters. Draw the characteristics of Low-pass filters 6

*** End ***